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(1) Research Program/Project/Study Title: Production and Utilization of Different Varieties of Sorghum as Feeds for Native Cattle and Darag Native Chicken	
Proponent Name:	Dr. Maryneth B. Barrios
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Program or Project has component (if no proceed to 2)? ☒ Yes ☒ No

Component Title:	Performance of Sorghum Varieties Using Chicken and Cow Manure as Fertilizer
Component Leader:	Dr. Aljen P. Vasquez
Academic Rank:	OIC Chief – DA ROS Astorga
Component Title:	Performance of Darag Native Chicken Fed with Sorghum Based Formulated Feeds Using Digital Application
Component Leader:	Dr. Maryneth B. Barrios
Academic Rank:	Asst. Prof. I
Component Title:	Performance of Native Cattle Fed with Different Varieties of Sorghum Silage
Component Leader:	Dr. Leah A. Ingalla
Academic Rank:	Asst. Prof. III

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Dumarao
Province:	Capiz
Region:	VI
(5) Classification Research: ☒ Basic ☐ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)
(7) Sector/Commodity Agriculture	(8) Discipline/Field of Specialization Native Animals

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(9) Rationale

The livestock sector accounts for 40% of the world's agriculture Gross Domestic Product (GDP). It employs 1.3 billion people, and creates livelihoods for one billion of the world's population living in poverty (FAO, 2006). However, this sector was affected due to COVID 19 Pandemic and due to global warming. Crop, livestock and fisheries have been affected by this pandemic. Food safety and security are the global concern at present scenario. The supply chain has been hit hardest by COVID-19, which causes food security of most vulnerable segment of population at risk (Poudel *et al.*, 2020). The Food and Agriculture Organization (FAO, 2020) states that COVID-2019 is affecting agriculture in two significant aspects: the supply and demand for food. Due to border restrictions, it affects the transportation of these agricultural products, inputs as well as man power. Aside from pandemic, another major threat for our livestock industry is the natural phenomenon that occurs in our environment brought by the climate change. The unpredictable period of rainy season, drought and even typhoon which severely affects the health status of many animals as well as the supply of feeds for this animals. This phenomenon as well as pandemic has brought a huge problem to our livestock owners because of the non stop increase of the inputs in crops such as fertilizer and feeds for livestock and poultry. With these, our government had made a lot of strategies on how to help our local farmers to increase their incomes and to sustain life by introducing lots of technologies in crop production as well as in livestock. One of these efforts is the introduction of the organic agriculture to farming both in crops and in livestock and planting of crops which are natural disaster resistant. In addition, Department of Agriculture also provided seeds to farmers to help them to grow their own food and to sustain their livelihood. One of the seeds that are now available in these government agency is the Sorghum which is considered as one of climate resistant plants.

Sorghum [*Sorghum bicolor* (L) Moench] is valued for its grain, stalks and leaves. It is one of the World's major cereal crops. Sorghum is a crop that stands out as an alternative to corn due to lower soil fertility demand and increased tolerance to drought. Sorghum grain has traditionally been used for livestock feed and stems and foliage for green chop, hay, silage, and pasture (Berenji and Dahlberg, 2004). It is also used as staple foods by some people in Africa and Asia as unfermented and fermented breads, porridges, couscous, rice-like products, snacks, and malted alcoholic and non-alcoholic beverages (Carson, *et al.*, 2020).

In the Philippines, sorghum is one of the considered primary component of feeds in livestock and poultry. This is now one of the major crops which the Department of Agriculture produced in order to be available to all Filipinos especially the livestock and poultry raisers as well as those engaged in feed industry. This is one of the major crops which are now prioritized by Department of Agriculture Research Outreach Station – Capiz at Brgy. Astorga, Dumarao, Capiz headed by Dr. Aljen P. Vasquez. Several varieties are now available at their research station and had already propagated and seeds are available for dispersal.

This study will be a collaborative research with DA ROS Astorga by using different varieties of sorghum and to be utilized as feeds for Darag native chicken and native cattle of the CapSU Dumarao Research Project and Dumarao Livestock Farm.

(9-A) Project Description

This project will be a collaborative study between Capiz State University, Bureau of Animal Industry and Department of Agriculture Research Outreach Station located in Brgy. Astorga, Dumarao, Capiz. The seeds will be provided by the Department of Agriculture and the project sites will be located at Dumarao Livestock Farm at Bungsuan, Dumarao, Capiz for native cattle and at Darag Native

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Chicken Research Project in CapSU Dumarao, College of Veterinary Medicine. Three studies will be conducted in this project where study 1 will be headed by Dr. Aljen Vasquez in the production of different sorghum varieties by using cattle and native chicken manure from the animal projects. The product of this study 1 will be used in study 2 and study 3 wherein the sorghum grains and stalks will be used as feeds for Darag native chicken and native cattle. Analysis of the samples such soil, manure and the feeds developed will be sent to DOST and or DA Feed Lab for analysis.

(10) Objectives of the Study

General Objective: Evaluate production of different sorghum varieties and sorghum based formulated feeds for native cattle and Darag native chickens.

Specific Objectives:

Study 1. Performance of Sorghum Varieties Using Chicken and Cow Manure as Fertilizer

- a. Evaluate the growth performance of different sorghum varieties using native cattle and chicken manure fertilizer.
- b. Determine the effect of native cattle and chicken manure fertilizer combination in the performance of different sorghum varieties.
- c. Identify the interaction effect between sorghum varieties and manure fertilizer.
- d. Find out the cost and return analysis of using native cattle and chicken as fertilizer.

Study 2. Performance of Darag Native Chicken Fed with Sorghum Based Formulated Feeds Using Digital Application

- a. Evaluate the growth performance of native chicken fed with different sorghum varieties based formulated feeds using digital application in terms of feed consumption, gain in weight, and feed conversion ratio.
- b. Determine the carcass quality of Darag native chicken fed with the formulated sorghum based feeds.
- c. Determine the shelf life of the formulated sorghum based feeds for native chickens.
- d. Find out the cost and return analysis of utilizing formulated sorghum based native chicken feeds.

Study 3. Performance of Native Cattle Fed with Different Varieties of Sorghum Silage

- a. Evaluate the growth performance of native cattle fed with different sorghum varieties silage.
- b. Compare the performance of native cattle fed with different varieties of sorghum.
- c. Determine the shelf life of the prepared sorghum silage.
- d. Find out the cost and return analysis of utilizing sorghum silage as feeds for cattle.

(11) Review of Literature/Prior Art Search

The proposed technology for using sorghum silage as feed to native cattle relates to the following registered technologies:

CN110558428A - Beef cattle fattening daily ration formula added with silage sorghum and application thereof. This invention from China provides a beef cattle fattening daily ration formula added with silage sorghum and application thereof. According to the invention, the optimum scheme of the daily ration formula for fattening the beef cattle is obtained through scientific blending of the daily ration formula in the feeding process of the sweet sorghum breeding cattle and scientific screening of the premix and the concentrate supplement, and scientific and reasonable feeding management rules for feeding the sweet sorghum breeding cattle are formulated at the same time, so that an ideal beef cattle fattening effect can be achieved, the comprehensive utilization efficiency of the sweet sorghum is improved, and the feeding cost is reduced. When the daily ration formula feeds the Ximenta cattle, test results show that the daily gain of a group of 50 percent of silage sorghum and 50

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percent of silage whole-plant corn is the best, the feed conversion efficiency is the highest, and the daily gain is 1.74 kg; the slaughtering performance of 50% of silage sorghum and 50% of silage whole-plant corn is high, the slaughtering rate is 57%, and the highest digestibility is 59.22%, so that the beef cattle meat yield can be increased by replacing part of silage whole-plant corn with silage sorghum.

On the other hand, proposed technology for utilization of sorghum as feeds to native chicken relates to these registered technologies:

CN104664069A - Compound additive of chicken sorghum feed. The invention discloses a compound additive of a chicken sorghum feed. The compound additive comprises the following components in parts by weight: 10-15 parts of lysine, 10-15 parts of methionine, 3-5 parts of threonine, 2-3 parts of tryptophan, 2-3 parts of valine, 7-13 parts of compound enzyme, 4-8 parts of a micro-ecological preparation, 20-25 parts of calcium hydrogen phosphate and 20-25 parts of montmorillonite. Through the compound additive provided by the invention, the utilization ratio of the sorghum feed is greatly increased, and the sorghum feed can replace maize feed to be fed to meat chickens.

CN104757295B - A kind of big chicken feed of wheat sorghum type yellow-feather broiler. The invention discloses a kind of big chicken feed of wheat sorghum type yellow-feather broiler, belong to poultry feed technical field, the relatively resistance to crude feed of yellow-feather broiler, particularly in the big chicken stage, it can largely use miscellaneous dregs of rice (such as cotton dregs, double-low rapeseed meal, palm kernel meal and corn germ cake) to replace dregs of beans in feed, reduce feed cost. Certain corn protein powder and corn germ cake are added in feed can meet the needs of yellow-feather broiler later stage coloring. Due to the wheat, the palm kernel meal etc. that are used in feed, addition xylanase eliminates the negative effect of the xylan in wheat, and addition mannanase eliminates the negative effect of the mannan in palm kernel meal. Protease utilization rate in sorghum is low, adds the digestive utilization ratio that certain protease improves amino acid. Sorghum and wheat, the additive amount relative increase of grease, the utilization rate of addition lipase raising grease are used due to a large amount of. This feed is no corn-soybean meal formula, can reach corn bean pulp type and be formulated identical feeding effect.

These technologies however does not discloses the breed of animals as well as the ingredients to be used in this study.

(12) Scientific Basis/Theoretical Framework

Early studies in pig and broiler chicken feeding experiments indicated sorghum-based diets to be inferior to corn-based diets in average daily gain, average daily feed intake, and feed per pound of gain (Ronda *et al.*, 2019).

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(13) Methodology

Study 1. Performance of Sorghum Varieties Using Chicken and Cow Manure as Fertilizer

This study will be conducted at Dumarao Livestock Farm. Prior to the start of the study, soil analysis will be taken from the area to assess the type of fertilizer to be used. This will be sent to the soil laboratory for analysis. Different varieties of sorghum seeds will be taken from DA ROS Astorga. Factorial experiment will be used in this study as follows:

Factor A: Sorghum Varieties

- | | |
|---------------------|-------------|
| A1 – Spv 422 Line 1 | A5 – Pop 2 |
| A2 – Spv 422 Line 2 | A6 – Pop 3 |
| A3 – Spv 422 Line 3 | A7 – Hybrid |
| A4 – Pop 1 | |

Factor B: Type of Fertilizer

- B1 – 100% Commercial Fertilizer
- B2 – 50% Commercial Fertilizer + 50% Native Cattle Manure
- B3 – 50% Commercial Fertilizer + 50% Native Chicken Manure
- B4 – 50% Native Cattle Manure + 50% Native Chicken Manure

The study will be laid out in Randomized Complete Block Design (RCBD) and study will be conducted in two cropping seasons.

Study 2. Performance of Native Chicken Fed with Different Sorghum Varieties Based Formulated Feeds Using Digital Application

This study will be conducted at Darag Native Chicken Research Project. Sorghum grains from study 1 will be used in this study for the formulation of the feeds of native chickens. Formulation of the feeds will use digital application from the mobile phones where all feed ingredients requirements will automatically analyzed to meet the desired crude protein needs of the native chickens. Feeds to be formulated will be growers feeds and this will also added with some other naturally occurring feedstuffs. Best three varieties among sorghum in study 1 will be used in this study. The experiment will be laid out in Completely Randomized Design (CRD) and will use a total of 120 heads of 2 months old Darag native chicken regardless of sex. The following will be the treatments of the study:

- Treatment 1 – 100% Commercial Feeds
- Treatment 2 – Spv 422 Variety Sorghum Feeds
- Treatment 3 – Pop Variety Sorghum Feeds
- Treatment 4 – Hybrid Variety Sorghum Feeds

Note: Other ingredients will also includes: Corn, Ricebran, Copra Meal, Azolla, Tricanthera, Multivitamins, and Antimolds agent.

Data to be gathered in this study will be the productive performance of chickens that will includes the average feed consumption, gain in weight and feed conversion ratio. For the carcass quality, data to be gathered will includes the slaughter weight, dressing weight, dressing percentage, major cuts and minor cuts weight and the color of skin, fats and muscles.

Study 3. Performance of Native Cattle Fed with Different Varieties of Sorghum Silage

This study will be conducted at Dumarao Livestock Farm of the Bureau of Animal Industry. All necessary documents between CapSU and BAI will be done prior to the start of the study. Best sorghum varieties from study 1 will be utilized in this study and will be laid in Completely Randomized Design with the following treatments:

- Treatment 1 – Spv 422 Variety
- Treatment 2 – Pop Variety

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Treatment 3 – Hybrid

All treatments will be replicated three times and with one cattle per replicate. Nine growing cattle of the same age will be used in this study. They will be fed with the prepared silage for the entire duration of the study (3 months). The data to be gathered in this study will be the average daily gain (ADG) of the cattle, digestibility of the dry matter (DM), organic matter (OM), non protein organic matter, and metabolizable energy and the carcass quality.

Laboratory Analysis: All samples that needs analysis will be collected and sent to specific laboratory for analysis. For soil sample, this will be sent to Soil laboratory in DA Regional Soils Laboratory for analysis; for the feeds samples, this will be sent to DA Regional Feed Laboratory and for the fecal samples for the digestibility study, this will be sent to specific laboratory that will analyse the samples.

Shelf Life Analysis. The produced feeds and silage will be analyzed for its shelf life. This will be done in a laboratory for a better result. This will be submitted to a laboratory that cater shelf life analysis.

Cost and Return Analysis. This will be done after the study is completed. All the expenses in the production of the feeds and silage will be computed and return of investment will also be computed.

Statistical Analysis. The resultss of the three studies will be analyzed using Analysis of Variance (ANOVA). Differences among treatment means will be compared using the Least Significant Difference (LSD). Results will be interpreted at five percent level of significance.

(14) Major Activities/Work plan

2024 - 2025

No	Description of activities	2024				2025			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1	Preparation of Documents								
2	Procurement of research materials								
3	Research implementation								
4	IP Protection Application								
5	IEC materials/Flyers production								
6	Technology Commercialization								
7	Extension of Technology to Community (Training)								

(15) Expected Output

EXPECTED OUTPUTS	DETAILS	TARGET
People Services	Train/ mentored native chicken growers and cattle growers group or association for the	100

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	utilization of sorghum as feeds.	
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials for utilization of sorghum as feeds as well as the sorghum based formulated feeds and silage making of sorghum.	2
Patents/IP	File 2 copyrights for the developed IEC materials and videos.	2
Places and Partnerships	Forge linkages and partnerships to LGU's, Capiz Darag Growers Association, Capiz Tourism Office, DTI, and other organizations.	5
Products	Produced low cost native chicken feeds and sustainbale sorghum based feeds for native chicken and cattle.	
Policy	Develop one policy inputs for sustainable sorghum production as feeds for livestock and poultry	1
Impact to Social/ Economics	<i>Social Impact:</i> • More available and low cost feeds for cattle and chicken • Technology transfer and commercialization strengthened • Strengthened Public-Private Partnerships • Innovation ecosystem improved • Technopreneurship empowered • Local agribusinesses supported <i>Economic Impact:</i> • Increase income of native chicken farmers	

(15) Technology Transfer Activity Design

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(16) Target Beneficiaries

- ✓ Native Chicken raisers
- ✓ Cattle raisers
- ✓ Department of Agriculture/ BAI/PNAD
- ✓ Local Government Units (Veterinary/ Livestock office)

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Dr. Maryneth B. Barrios	Leader	Veterinary Medicine/Animal Science
Dr. Leah A. Ingalla	Member	Veterinary Medicine/Animal Science
Dr. Aljen P. Vasquez	Member	Agronomy
Dr. Emely Escala		Nutrition

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (CapSU)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	2 @ 11,068 x 24 months	531,264.00
Maintenance and Other Operating Expenses (MOOE)		
Transportation allowance		30,000
Representation Expenses		30,000
Training Expenses		50,000
Supplies (Agricultural and Veterinary)		100,000
Communication		5,000
Laboratory Analysis		30,000
CAPITAL OUTLAY		
Shredder Machine		100,000
Repair of Animal Shed and fences		200,000

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